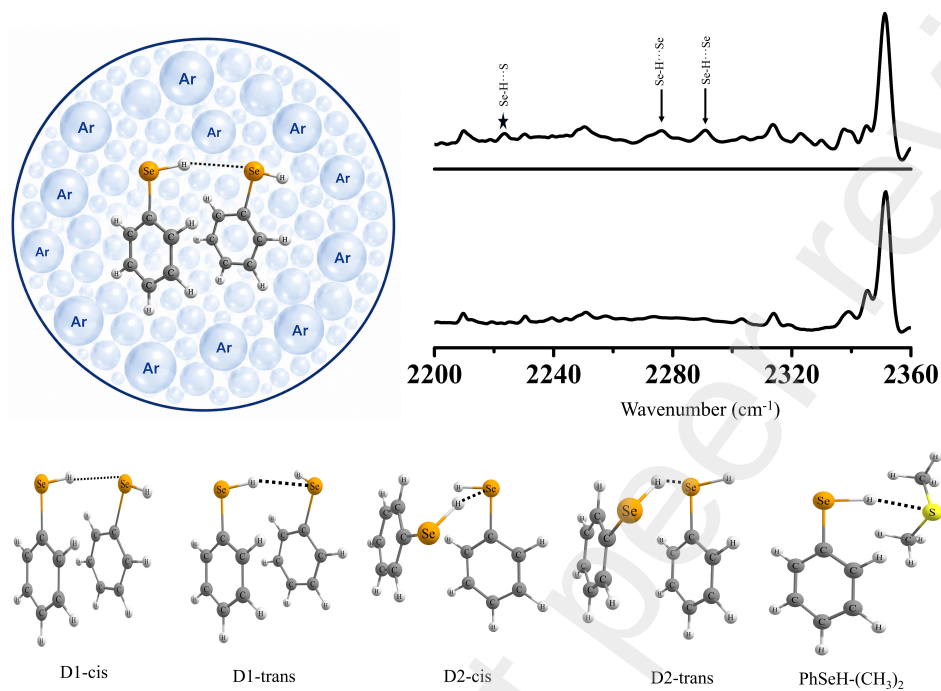


Graphical Abstract

Matrix-Isolation Infrared Spectroscopic Evidence of Se–H...Se and Se–H...S Hydrogen Bonding in Benzeneselenol Complexes

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Highlights

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- Spectroscopic evidence of Se–H...Se and Se–H...S hydrogen bonds.
- Se-H stretching bands red-shift by 75, 61, and 128 cm⁻¹.
- The benzeneselenol dimers are dispersion-driven.
- The benzeneselenol-dimethyl sulfide complex is electrostatic-driven.

Matrix-Isolation Infrared Spectroscopic Evidence of Se–H...Se and Se–H...S Hydrogen Bonding in Benzeneselenol Complexes

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ARTICLE INFO

Keywords:

Selenium
Sulphur
Hydrogen bonding
Benzeneselenol
Infrared spectroscopy

ABSTRACT

The hydrogen-bond donor ability of selenium has been investigated by studying benzeneselenol dimers and the complex of benzeneselenol with dimethyl sulfide ($\text{S}(\text{CH}_3)_2$) using matrix-isolation infrared spectroscopy and electronic-structure calculations. The structures and relative stability of the complexes were characterised computationally, while their spectroscopic signatures were examined under cryogenic Ar-matrix conditions. Formation of $\text{Se}-\text{H} \cdots \text{Se}$ and $\text{Se}-\text{H} \cdots \text{S}$ hydrogen bonds was established from the characteristic red shift of the $\text{Se}-\text{H}$ stretching vibration mode of benzeneselenol, supported by comparison with simulated vibrational spectra. Several benzeneselenol dimer conformers with closely spaced stabilisation energies were predicted. The calculated vibrational frequencies of these four conformers were assigned to the experimentally observed $\text{Se}-\text{H}$ stretching bands at 2276 and 2290 cm^{-1} . These bands correspond to a red shift of 75 and 61 cm^{-1} compared to the benzeneselenol monomer, and are assigned to the trans and cis conformer pairs, respectively. For the $[\text{PhSeH}-\text{S}(\text{CH}_3)_2]$ complex, a new band appeared at 2223 cm^{-1} upon annealing, corresponding to a red shift of 128 cm^{-1} , confirming the formation of a $\text{Se}-\text{H} \cdots \text{S}$ hydrogen bond. This work provides the first impression of the $\text{Se}-\text{H} \cdots \text{Se}$ hydrogen bond interaction in benzeneselenol dimer and the $\text{Se}-\text{H} \cdots \text{S}$ hydrogen bond interaction in $[\text{PhSeH}-\text{S}(\text{CH}_3)_2]$.

1. Introduction

Hydrogen bonding is one of the most important non-covalent interactions and plays a fundamental role in chemistry, biology, supramolecular chemistry, interstellar chemistry, and biophysics [1–9]. In the chalcogen series, oxygen- and sulfur-centred hydrogen bonding interactions have been extensively investigated, where oxygen and sulfur act both as hydrogen bond donors and acceptors [10–17]. In contrast, selenium-centred hydrogen bonding interactions remain comparatively less explored despite the growing importance of selenium-containing compounds in chemistry and biology [18, 19]. Selenocysteine, often referred as the 21st amino acid, contains a selenol ($\text{Se}-\text{H}$) functional group and is present in several human proteins [20, 21]. The $\text{Se}-\text{H}$ functional group is also present in selenomethionine. Organoselenium compounds exhibit remarkable versatility and participate in a wide range of transformations with applications extending from organic synthesis to materials chemistry and biology [18, 19]. Selenium-containing compounds have also attracted considerable attention because of their biological activities, including anticancer, antibacterial, antiviral, and antioxidant properties [20, 22, 23]. Owing to the unique electronic properties of selenium, selenium chemistry continues to emerge as an active area of investigation in biochemistry, ligand chemistry, and semiconductor science [18, 19, 24]. Understanding selenium-centred non-covalent interactions is therefore important for elucidating the structure, stability, and reactivity of selenium-containing molecular systems.

Recent investigations have primarily focused on the hydrogen bond acceptor ability of selenium. Several experimental investigations have established the formation of hydrogen bonds in which selenium serves as a hydrogen-bond acceptor in various molecular complexes. The first experimental evidence for $\text{O}-\text{H} \cdots \text{Se}$ hydrogen bonding interaction was reported using molecular jet spectroscopy experiments [25] with phenol as the hydrogen bond donor and $\text{Se}(\text{CH}_3)_2$ as the hydrogen bond acceptor. The complexation-induced red shift in the OH stretching frequency ($\Delta\nu_{\text{OH}}$) of phenol was found to be 240 cm^{-1} . More recently, hydrogen-bonded complexes of alcohols with $\text{Se}(\text{CH}_3)_2$ were investigated using room-temperature gas-phase FTIR spectroscopy [26]. The observed $\Delta\nu_{\text{OH}}$ values for CH_3OH and $\text{C}_2\text{H}_5\text{OH}$ were 113 and 124 cm^{-1} , respectively. Interestingly, nearly identical red shifts were observed for a given donor molecule with $\text{O}(\text{CH}_3)_2$ and $\text{S}(\text{CH}_3)_2$, suggesting that selenium exhibits hydrogen-bond acceptor ability comparable to that of oxygen

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and sulfur in these systems. Recent matrix-isolation infrared spectroscopic and theoretical investigations on [ROH–Se(CH₃)₂] complexes (R = H, CH₃, and C₂H₅) also demonstrated that selenium can act as an efficient hydrogen bond acceptor with hydrogen bond strengths comparable to those involving oxygen and sulfur [27]. The C–H...X hydrogen bonding interactions in [CHCl₃–X(CH₃)₂] complexes (X = O, S, and Se) have been investigated using matrix-isolation infrared spectroscopy and electronic structure calculations [28]. The observed red shift in the $\nu_{\text{C-H}}$ stretching vibration for the selenium-containing complex was found to be comparable to that of the sulfur analogue, suggesting that selenium can act as an efficient hydrogen bond acceptor. Matrix-isolation infrared spectroscopic investigations on [(CH₃)₂NH–Se(CH₃)₂] further established the formation of N–H...Se hydrogen bonding interaction [29]. In addition, comparative investigations involving ammonia complexes with O(CH₃)₂, S(CH₃)₂, and Se(CH₃)₂ revealed that the selenium-containing complex exhibits the largest red shift in the N–H stretching frequencies [30]. Hence, the Se–H group of selenols is expected to act as an efficient hydrogen-bond donor.

Selenols are more acidic than the corresponding thiols owing to the weaker Se–H bond and the greater size and polarizability of selenium, which provide enhanced stabilisation of the resulting selenolate anion. [31] However, information on selenium-centred hydrogen-bonding (SeCHB) interactions, in which selenium serves as the hydrogen-bond donor, remains limited. One of the earliest investigations of intermolecular SeCHB interactions was reported for the H₂Se–HF complex. The formation of Se–H...F hydrogen-bond interaction was confirmed by matrix-isolation IR spectroscopy in Ar matrix at 12 K [32]. A theoretical investigation on intermolecular SeCHB interactions involved in the dimers of [H₂Se–YH₂] (Y = O, S, and Se), where the structures and stabilisation energies of the hydrogen-bonded complexes were investigated using DFT and MP2 calculations [33]. The complexes of para-substituted pyridines with SeH₂ have been investigated theoretically to understand the influence of electron-donating and electron-withdrawing substituents (NH₂, OH, CH₃, H, F, Cl, CN, NO₂) on the hydrogen bond parameters [34]. The study demonstrated that electron-donating substituents at the para position of pyridine enhance the interaction strength of the Se–H...N hydrogen-bond. The calculated stabilisation energies of the Se–H...N interactions were found to lie in the range of 10–15 kJ/mol, accompanied by significant red shifts of 122–149 cm^{–1} in the Se–H stretching frequency. Theoretical studies on selenocysteine have identified several low-energy conformers stabilised by intramolecular Se–H...O and Se–H...N hydrogen-bond interactions [35]. Formation of intramolecular Se–H... π hydrogen-bond interaction has been identified in 3-buteneselenol [36], 3-butyne-1-selenol [37], 2-Propene-1-selenol [38], and propargyl selenol [39], while investigating the most stable conformer of the molecules using microwave spectroscopy. In addition to spectroscopic and theoretical studies, crystallographic investigations have provided evidence for SeCHB, in which selenium acts as a hydrogen-bond donor. The crystal structures of thiophenol and selenophenol have been investigated, demonstrating the existence of intermolecular S–H...S and Se–H...Se hydrogen-bond interactions [40]. Recently, Se...Se interaction has been investigated in the crystals of organoselenium compounds with the help of electronic structure calculations. The presence of Se–H...Se hydrogen-bond interaction has been proposed from non-covalent interaction (NCI) analysis. [41] Collectively, these studies establish that selenium can act effectively as a hydrogen-bond donor and that the strength of these interactions is strongly influenced by selenium's large polarizability and diffuse valence electron density. Despite these theoretical and crystallographic investigations, direct spectroscopic evidence for selenium-centred hydrogen-bonding interactions in which selenium acts as a hydrogen-bond donor remains lacking. To the best of the author's knowledge, no direct experimental evidence of Se–H acting as a donor in Se–H...Se and Se–H...S motifs is available.

Herein, Se–H...Se and Se–H...S hydrogen-bond interactions have been studied by experimental and theoretical investigations of benzeneselenol (PhSeH) in dimer ((PhSeH)₂) and [PhSeH–S(CH₃)₂] complex using matrix-isolation IR spectroscopy and electronic structure calculations. The equilibrium structure of benzeneselenol has recently been investigated using rotational spectroscopy and quantum chemical calculations providing important structural insight into relatively less explored selenium-containing aromatic systems [42]. Unlike phenol and thiophenol, both of which possess essentially planar equilibrium structures, the equilibrium structure of benzeneselenol is non-planar due to the torsional flexibility associated with the Se–H group. At the B3PW91/6-311++G(d,p) level of theory, the hydrogen atom attached to selenium deviates by approximately 12° from the molecular plane. For the first time, experimental IR spectra of (PhSeH)₂ and [PhSeH–S(CH₃)₂] have been recorded corresponding to the Se–H stretching vibration ($\nu_{\text{Se-H}}$) under matrix-isolation conditions. Moreover, the complexation-induced shift in $\nu_{\text{Se-H}}$ mode ($\Delta\nu_{\text{Se-H}}$) due to the formation of Se–H...Se and Se–H...S hydrogen-bond interactions have been estimated. Experimental IR spectra of (PhSeH)₂ and [PhSeH–S(CH₃)₂] have been assigned to the plausible conformers based on the electronic structure calculations. The nature and origin of Se–H...Se and Se–H...S hydrogen-bond interactions have been

further analysed using energy decomposition analysis (EDA), natural bond orbital (NBO) analysis, and atoms in molecules (AIM) analysis.

2. Experimental and Computational details

All the experiments were performed under matrix-isolation conditions and analysed using Fourier-transform infrared (FTIR) spectroscopy. Details of the experimental setup have been described elsewhere [43]. Benzeneselenol (PhSeH, of purity >97%) and dimethyl sulfide (S(CH₃)₂), of purity 99%) were procured from Sigma-Aldrich and used without further purification. High-purity argon gas (99.999%) was premixed with vapours of PhSeH and S(CH₃)₂ in a 1 L stainless steel mixing chamber. Several gas-phase concentrations were initially examined to optimise the spectral quality, minimise self-association of the monomers, and maximise the signal-to-noise ratio. For (PhSeH)₂ experiments, optimal experimental conditions were achieved for a 1:2000 mixture of PhSeH:Ar. For the experiments of [PhSeH-S(CH₃)₂] complex, optimal conditions were obtained for 2:1 mixture of PhSeH:S(CH₃)₂ with dilution ratios with Ar of approximately 1:3000 and 1:6000 with respect to PhSeH and S(CH₃)₂, respectively. The prepared gas mixtures were deposited through an effusive nozzle (1.58 mm diameter) onto a polished KBr window maintained at 16 K using a closed-cycle helium cryostat. Sample deposition was carried out using a dosing valve at approximately 46 mmol h⁻¹ for 180 minutes and 210 minutes for PhSeH and [PhSeH-S(CH₃)₂] mixture, respectively. The argon matrix containing isolated PhSeH was annealed for 120 min to facilitate the formation of (PhSeH)₂, whereas PhSeH-S(CH₃)₂ matrix was annealed for 60 min to promote the formation of [PhSeH-S(CH₃)₂] complex. Annealing was carried out by gradually increasing the temperature of the matrix from 16 to 35 K, after which the matrix was cooled back to 16 K, and the infrared spectra were recorded again. All FTIR spectra were recorded at 16 K using a Tensor II spectrometer (Bruker Optics) equipped with a DLaTGS detector and a KBr broadband beamsplitter. The full IR spectrum of PhSeH has been presented in ESI Figure S1. The spectra were recorded in the range of 400–4000 cm⁻¹ with an instrumental resolution of 1 cm⁻¹. The spectrometer and sample compartment were continuously purged with dry nitrogen gas to eliminate spectral interference from atmospheric CO₂ and H₂O.

Electronic structure calculations for the monomers and the corresponding complexes were performed using the Gaussian 16 programme package [44] using the PBE-D3 functional in conjunction with 6-311++G(2d,2p) level. All the coordinates were fully relaxed to scan the dimer potential energy surface of (PhSeH)₂ and [PhSeH-S(CH₃)₂] complex. Harmonic vibrational frequency calculations confirmed that all optimised structures correspond to minima on the potential energy surface. The binding energy (BE) of each complex was calculated as the difference between the energy of the complex and the sum of the energies of the isolated monomers. The binding energies were further corrected for harmonic zero-point vibrational energy (ZPVE) and basis set superposition error (BSSE) using the counterpoise method [45], which yields the corresponding stabilisation energies (D₀). The stabilisation energies of the complexes are listed in ESI Table S1 along with all the energy values. Quantum theory of atoms in molecules (QTAIM) and natural bond orbital (NBO) analyses were performed using the AIMAll [46] and NBO 7.0 [47] program packages, respectively, to characterise the nature of the non-covalent interactions in the complexes. The second-order perturbation stabilisation energy ($\Delta E_{ij}^{(2)}$) associated with charge-transfer interactions was obtained from NBO analysis. Symmetry-adapted perturbation theory calculations at the SAPT2 [48] level were performed using the PSI4 program package [49] to partition the total interaction energy of the complexes into electrostatic, induction, dispersion, and exchange-repulsion contributions.

3. Results and discussion

Electronic structure calculations have been performed to manually scan the potential energy surface of (PhSeH)₂ dimer and 1:1 complex of [PhSeH-S(CH₃)₂]. The results of theoretical calculations are described first, followed by those of the matrix isolation IR spectroscopy experiment. Finally, experimental IR spectra have been assigned to the corresponding hydrogen-bonded conformers using calculated geometries and simulated vibrational frequencies.

3.1. Possible structures of (PhSeH)₂ and [PhSeH-S(CH₃)₂] complexes:

The PhSeH monomer exhibits conformational flexibility along the ∠C–C–Se–H torsional coordinate, with minima obtained at different dihedral angles. The energy differences between these minima are very small, indicating a shallow potential energy surface along this coordinate. The corresponding dihedral angles and relative energies of the monomer minima are summarised in ESI Table S2.

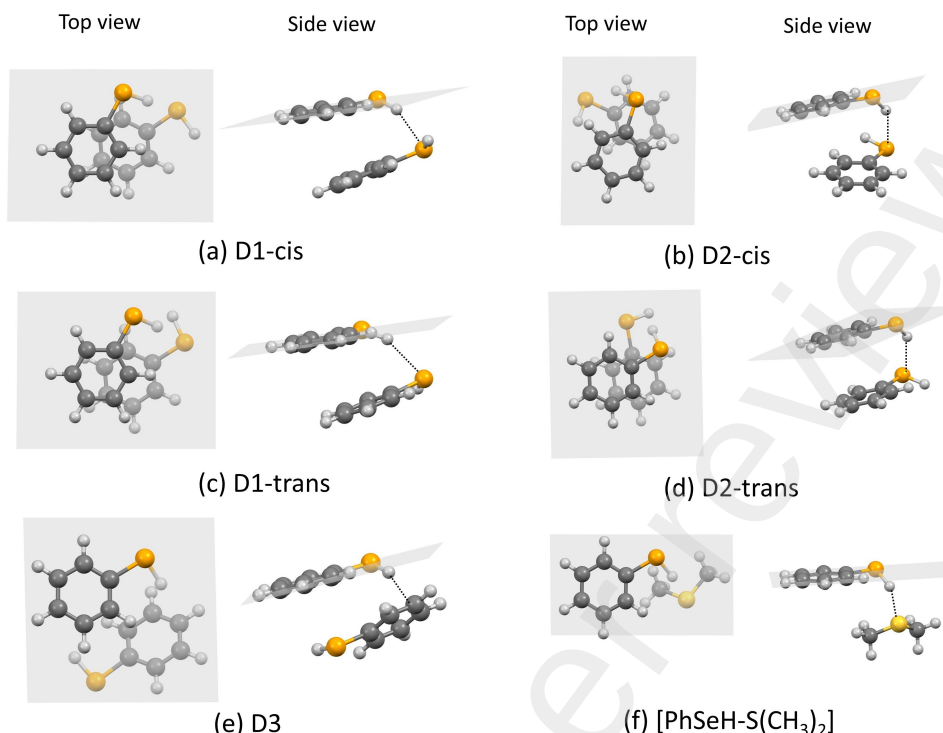


Figure 1: Optimized geometries of D1-cis (a), D2-cis (b), D1-trans (c), D2-trans (d), D3 (e), and [PhSeH-S(CH₃)₂] (f) obtained at the PBE-D3/6-311++G(2d,2p) level of theory. Top and side views of each structure are shown.

Electronic structure calculations on the (PhSeH)₂ potential energy surface resulted in five distinct minima, designated as D1-cis, D2-cis, D1-trans, D2-trans, and D3, together with one minimum corresponding to [PhSeH-S(CH₃)₂] complex. The nomenclature of cis and trans has been introduced for the (PhSeH)₂ to represent the relative orientation of the Se–H groups of the two PhSeH monomers. The optimised geometries of the complexes are shown in Figure 1, and the corresponding geometrical parameters, calculated at the PBE-D3/6-311++G(2d,2p) level, are summarised in Table 1. The dimer structures, D1 and D2, are stabilised by Se–H...Se hydrogen bond and $\pi - \pi$ interactions with stabilisation energy (D_0) in the range of 23.24 - 24.57 kJ/mol. The most stable D2-trans is characterised by hydrogen bond length ($R_{H...Se}$) of 2.812 Å along with hydrogen bond angle ($\angle\text{Se–H–Se}$) of 143°. The $R_{H...Se}$ and $\angle\text{Se–H–Se}$ in the next local minimum, D1-trans, are 2.834 Å and 154°, respectively. The next local minimum D2-cis, exhibits $R_{H...Se}$ and $\angle\text{Se–H–Se}$ of 2.859 Å and 147°, respectively, while the corresponding values for D1-cis are 2.821 Å and 154°, respectively. Formation of the hydrogen bond results in elongation of the Se–H bond and corresponding red shift in the $\nu_{\text{Se–H}}$ mode. For the D1 and D2, the Se–H bond elongations are within the range of 0.008 - 0.009 Å compared to the monomer. The corresponding red-shift in the $\nu_{\text{Se–H}}$ mode is 90 (D2-cis), 94 (D1-cis), 100 (D2-trans) and 101 (D1-trans) cm⁻¹.

The other minimum, D3, is stabilised by Se–H... π hydrogen bond interaction involving the Se–H group of one PhSeH molecule and the π -electron density of the second benzeneselenol molecule. In addition, weak C–H... π interaction contributes to the stabilisation of the complex. The distance between the hydrogen atom of the Se–H group and the interacting ring carbon atom of the second PhSeH molecule is 2.784 Å. The D_0 of D3 is calculated to be 21.05 kJ/mol. Formation of the Se–H... π hydrogen bond interaction leads to elongation of the Se–H bond length of the donor PhSeH molecule and results in a comparatively smaller red shift of 17 cm⁻¹ in the $\nu_{\text{Se–H}}$ mode compared to the monomer.

The optimised geometry of [PhSeH-S(CH₃)₂] complex is stabilised by Se–H...S hydrogen-bond interaction. The hydrogen bond length ($R_{H...S}$) is calculated to be 2.366 Å, while the corresponding $\angle\text{Se–H–S}$ hydrogen-bond angle is 156°. Formation of the hydrogen bond leads to an elongation of the Se–H bond length of the donor PhSeH molecule

Table 1

Stabilisation energy (D_0 , kJ/mol), hydrogen bond length ($R_{H...X}$ with $X = S$ and Se , in Å), hydrogen bond angle (θ in °), C–C–Se–H dihedral angle for donor (D_d) and acceptor (D_a) (in °) Se–H bond length (R_{Se-H} , in Å), change in Se–H bond length (ΔR_{Se-H} , in Å), ν_{Se-H} (in cm^{-1}), and change in ν_{Se-H} ($\Delta \nu_{Se-H}$, in cm^{-1}) for three conformers of $(PhSeH)_2$ and $[PhSeH-S(CH_3)_2]$ obtained at the PBE-D3/6-311++G(2d,2p) level of theory. D3 involves a $Se-H \cdots \pi$ interaction.

Complex	D_0	$R_{H...X}$	θ	D_d	D_a	R_{Se-H}	ΔR_{Se-H}	$\Delta \nu_{Se-H}$
D1-cis	23.24	2.821	154	23	15	1.489	0.008	94
D2-cis	23.43	2.859	147	44	8	1.490	0.008	90
D1-trans	23.85	2.834	154	23	21	1.489	0.009	101
D2-trans	24.57	2.812	143	45	28	1.491	0.009	100
D3	21.05	^a	^b	23	28	1.482	0.001	17
$[PhSeH-S(CH_3)_2]$	24.19	2.366	156	34	^c	1.509	0.028	289

^aHydrogen to carbon distance is 2.784 Å; ^b $\angle Se-H-C$ is 155°; ^cNot applicable

from 1.481 Å in the monomer to 1.509 Å in the complex. The stabilisation energy (D_0) of the complex is calculated to be 24.19 kJ/mol. The stabilisation energy of the complex is comparable to those calculated for the benzeneselenol dimers, indicating the formation of a strong $Se-H \cdots S$ hydrogen-bond interaction. The elongation of the Se–H bond is accompanied by a pronounced red shift in the ν_{Se-H} mode, with calculated shift of 289 cm^{-1} compared to the monomer, consistent with the formation of a strong hydrogen-bond interaction

3.2. Nature of Interactions in $(PhSeH)_2$ and $[PhSeH-S(CH_3)_2]$ complexes:

QTAIM analysis was performed to investigate the nature of the intermolecular interactions present in $(PhSeH)_2$ and $[PhSeH-S(CH_3)_2]$ complex. The molecular graphs of the optimised structures, along with the corresponding bond critical points (BCPs), are presented in Figure 2. The values of electron density (ρ) and Laplacian ($\nabla^2\rho$) of electron density at the BCP are presented in ESI Table S3. The molecular graph of D1-cis exhibits a (3,-1) bond critical point (BCP) corresponding to the $Se-H \cdots Se$ hydrogen bond interaction, along with BCPs corresponding to secondary $\pi-\pi$ interaction. The ρ and $\nabla^2\rho$ at the BCP associated with the $Se-H \cdots Se$ interaction are 0.005 and 0.013 au, respectively. The presence of the additional BCP corresponding to the secondary $\pi-\pi$ interaction between the carbon atoms of both rings ($\rho = 0.005$ au, $\nabla^2\rho = 0.015$ au) further supports the cooperative stabilisation of the dimer structure. For D2-cis, the molecular graph reveals the presence of a bond path corresponding to the $Se-H \cdots \pi$ interaction between the hydrogen atom of the Se–H group and the aromatic ring carbon atom of the acceptor molecule. The corresponding values of ρ and $\nabla^2\rho$ are 0.006 and 0.016 au, respectively. In addition, a BCP corresponding to the $Se-H \cdots Se$ hydrogen bond interaction is observed, with ρ and $\nabla^2\rho$ values of 0.011 and 0.023 au, respectively, indicating that $Se-H \cdots Se$ hydrogen bond interaction remains the dominant stabilising contact within the dimer. The molecular graph of D1-trans also shows a (3,-1) bond critical point between the hydrogen atom of the Se–H group and the selenium atom of the acceptor molecule. The values of ρ and $\nabla^2\rho$ at the corresponding BCP are 0.011 and 0.023 au, respectively, comparable to those obtained for the D1-cis and D2-cis conformer. A secondary $\pi-\pi$ interaction is also identified, with ρ and $\nabla^2\rho$ values of 0.005 and 0.015 au, respectively. For D2-trans, the molecular graph shows a (3,-1) bond critical point corresponding to the $Se-H \cdots Se$ hydrogen-bond interaction alone, with ρ and $\nabla^2\rho$ values of 0.012 and 0.025 au, respectively. The D3 is characterised by the presence of $Se-H \cdots \pi$ interaction between the hydrogen atom of the Se–H group and an aromatic ring carbon atom of the acceptor molecule. The corresponding values of ρ and $\nabla^2\rho$ are 0.008 and 0.020 au, respectively.

The molecular graph of the $[PhSeH-S(CH_3)_2]$ complex obtained from AIM analysis is shown in Figure 2. The $Se-H \cdots S$ hydrogen-bond interaction between the hydrogen atom of the donor Se–H group and the sulfur atom of $S(CH_3)_2$ is characterised by the presence of (3,-1) BCP. The values of ρ and $\nabla^2\rho$ at the corresponding BCP are 0.025

and 0.044 au, respectively, indicating a strong Se-H...S hydrogen-bond interaction in [PhSeH-S(CH₃)₂] complex compared to the (PhSeH)₂, consistent with the greater proton-accepting ability of the sulfur in S(CH₃)₂.

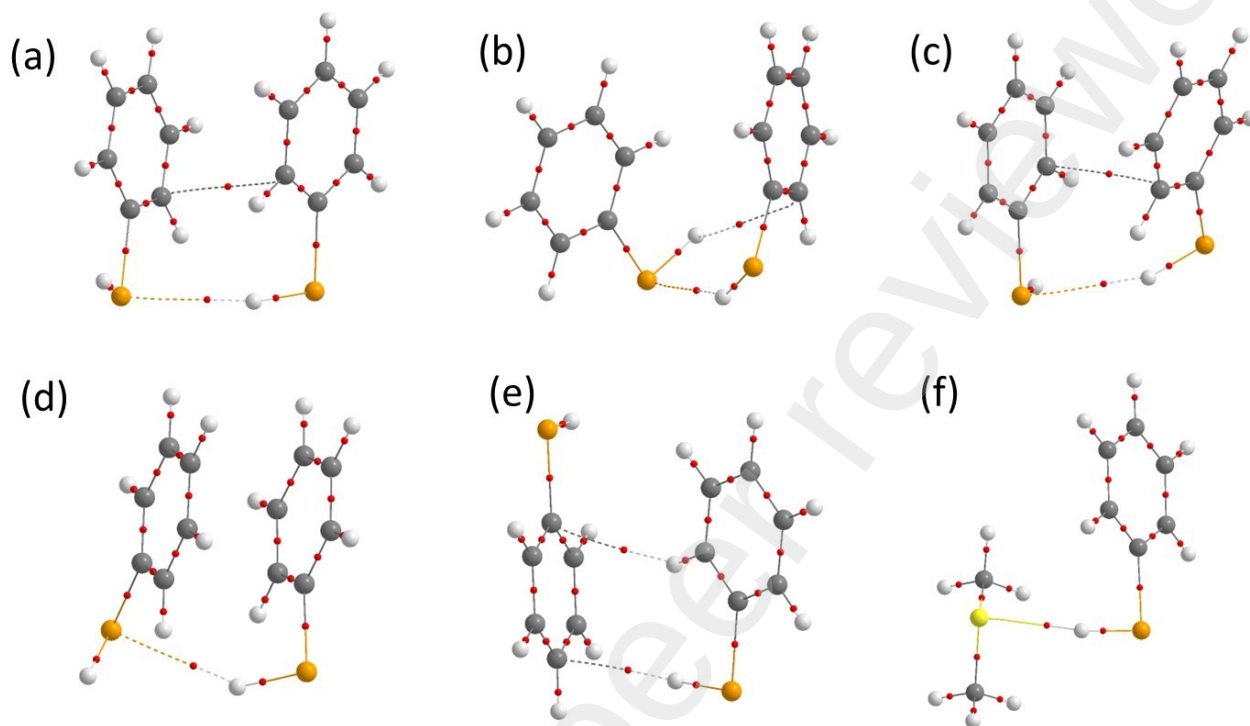


Figure 2: QTAIM plot of D1-cis (a), D2-cis (b), D1-trans (c), D2-trans (d), D3 (e), and [PhSeH-S(CH₃)₂] (f) obtained at the PBE-D3/6-311++G(2d,2p) level of theory. The values of ρ and $\nabla^2\rho$ corresponding to the bond critical points are presented in ESI Table S3. The non-CP Threshold was set to 0.005 au, and only critical points with $\rho > 0.005$ au are displayed.

NBO analysis was performed to elucidate the origin of the noncovalent interactions. The donor NBO, acceptor NBO, and the corresponding donor-acceptor interacting NBOs are presented in Figure 3, whereas the total charge transfer and second-order perturbation energies ($\Delta E_{ij}^{(2)}$) are summarised in ESI Table S4. The Se-H...Se hydrogen-bond interactions in the D1 and D2 dimers are characterised by charge transfer from the lone pair orbital of the acceptor selenium atom (LP_{Se}) to the antibonding σ_{Se-H}^* orbital of the donor PhSeH molecule. The total charge transfer in D1 and D2 lies in the range of 23.82 - 27.30 me, with corresponding charge-transfer energies ($\Delta E_{ij}^{(2)}$) of 14.27 - 17.28 kJ/mol. The corresponding change in the LP_{Se} and σ_{Se-H}^* occupancies are in the range of 15.65 - 24.10 me and 24.08 - 28.75 me, respectively, indicating appreciable donor-acceptor interaction associated with the Se-H...Se hydrogen bond interaction. The NBO analysis of D3 indicates a comparatively weaker charge-transfer interaction. The total charge transfer is 6.05 me with a corresponding $\Delta E_{ij}^{(2)}$ of 2.09 kJ/mol. The extent of the change in LP_{Se} and σ_{Se-H}^* occupancies is consistent with the weaker Se-H...Se interaction.

The [PhSeH-S(CH₃)₂] complex exhibits a substantially larger charge-transfer interaction compared to the (PhSeH)₂. The interaction is characterised by charge transfer from the sulfur lone pair of S(CH₃)₂ to the antibonding σ_{Se-H}^* orbital of PhSeH. The total charge transfer is 84.62 me, with a corresponding $\Delta E_{ij}^{(2)}$ of 51.76 kJ/mol. The changes in the occupancies of LP_{Se} and σ_{Se-H}^* are 76.55 and 77.10 me, respectively, representing a significantly strong donor-acceptor interaction in the Se-H...S hydrogen-bonded complex compared to Se-H...Se complexes.

Calculations using symmetry-adapted perturbation theory of second order (SAPT2) were performed to partition the total interaction energies of the complexes into their individual attractive components, namely, electrostatic (ΔE_{Elec}), induction (ΔE_{Ind}), and dispersion (ΔE_{Disp}) contributions. The SAPT2 interaction energy decomposition for the benzeneselenol dimers (D1, D2, and D3) and [PhSeH-S(CH₃)₂] complex calculated using aug-cc-pVDZ basis set is

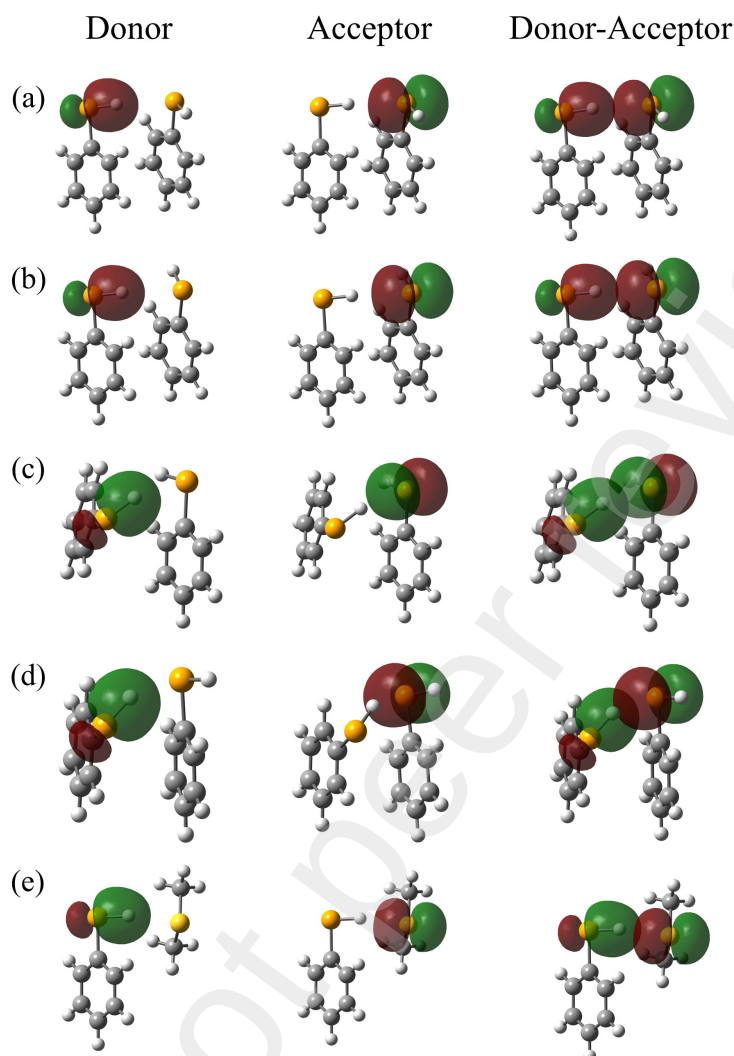


Figure 3: The donor NBO, acceptor NBO, and donor-acceptor interacting NBOs for the D1-cis (a), D2-cis (b), D1-trans (c), D2-trans (d), and the $[\text{PhSeH-S}(\text{CH}_3)_2]$ (e), obtained at the PBE-D3/6-311++G(2d,2p) level of calculation. The isovalue used for the surfaces is 0.04.

summarised in Figure 4 and ESI Table S5. The electrostatic component is the dominant contributor to the stabilisation of the $[\text{PhSeH-S}(\text{CH}_3)_2]$ complex, accounting for approximately 41% of the total attractive interaction energy. In contrast, the electrostatic contribution for dimers D1, D2, and D3 lies in the range of 28 - 30%, indicating comparatively weaker electrostatic stabilisation in the benzeneselenol dimers. The induction contribution for D1, D2, and D3 is relatively smaller and varies between 9% and 11% of the total stabilisation energy, whereas a significantly larger induction contribution of approximately 20% is obtained for the $[\text{PhSeH-S}(\text{CH}_3)_2]$ complex. The enhanced induction contribution in the $\text{Se-H} \cdots \text{S}$ hydrogen-bonded complex is consistent with the larger charge-transfer interaction observed from NBO analysis. Dispersion interactions also play an important role in stabilising the benzeneselenol dimers. The dispersion contribution for D1, D2, and D3 ranges from 61 to 63%, highlighting the importance of weak intermolecular interactions involving the aromatic rings in these complexes. For the $[\text{PhSeH-S}(\text{CH}_3)_2]$ complex, the dispersion contribution is comparatively smaller ($\sim 39\%$), although it still contributes significantly toward stabilisation of the complex. Overall, the SAPT analysis indicates that the benzeneselenol dimers are predominantly dispersion-driven, whereas the stronger $\text{Se-H} \cdots \text{S}$ hydrogen bond in the $[\text{PhSeH-S}(\text{CH}_3)_2]$ complex possesses a comparatively larger electrostatic and induction character.

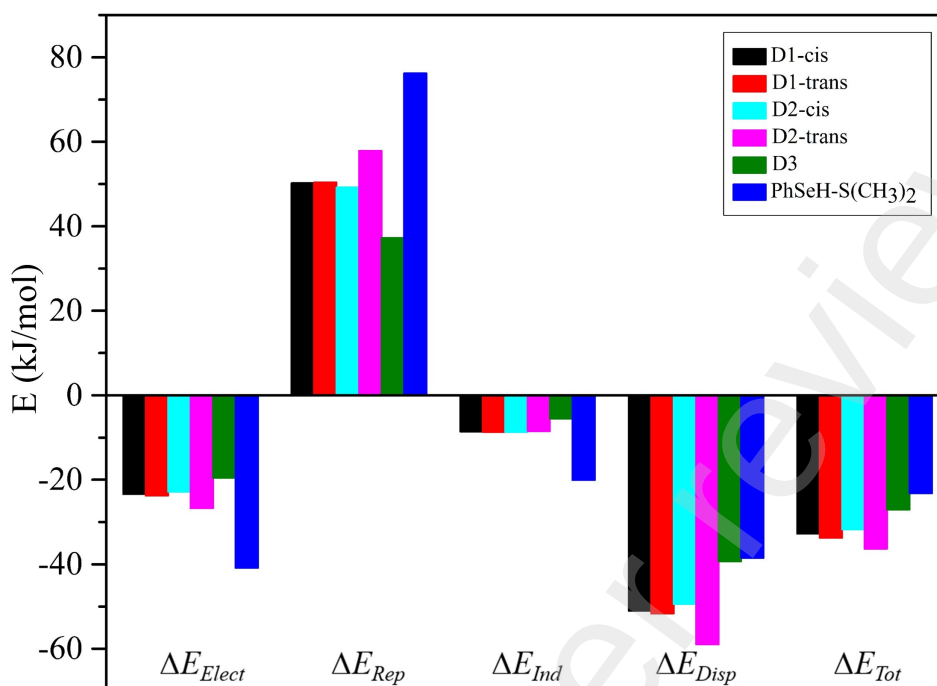


Figure 4: SAPT2/aug-cc-pVDZ decomposition of the interaction energy for the D1-cis, D2-cis, D1-trans, D2-trans, D3, and the [PhSeH-S(CH₃)₂]. The different components contributing to the total interaction energy (ΔE_{Tot}) in the SAPT2 method are electrostatics (ΔE_{Elect}), repulsion (ΔE_{Rep}), Induction (ΔE_{Ind}), and Dispersion (ΔE_{Disp}).

3.3. IR spectra of (PhSeH)₂ and [PhSeH-S(CH₃)₂] complexes in Ar matrix:

Before discussing the IR spectra of the hydrogen-bonded complexes, it is important to establish the assignment of the ν_{Se-H} mode of isolated PhSeH. Experimental spectroscopic information on PhSeH is limited, and high-resolution infrared spectra of isolated PhSeH either in the gas phase or under cryogenic conditions have not yet been reported. One of the experimental infrared spectroscopic studies of PhSeH was carried out on solid state KBr and PhSeH adsorbed on a gold surface [50]. A broad transition centred at 2305 cm⁻¹ observed in solid state KBr was assigned to the ν_{Se-H} mode of PhSeH. This transition was absent in the IR spectrum of PhSeH adsorbed on the gold surface. In the present work, the infrared spectrum of isolated PhSeH recorded under argon matrix-isolation conditions exhibits a distinct band at 2351 cm⁻¹ (Figure 5), which is assigned to the ν_{Se-H} mode. The assignment is further supported by electronic structure calculations and the characteristic frequency range expected for aromatic selenols. The monomer ν_{Se-H} band serves as the reference for identifying the complexation-induced shifts associated with the formation of Se–H···Se and Se–H···S hydrogen-bonded complexes.

After annealing the argon matrix, two new bands appear at 2276 and 2290 cm⁻¹ (Figure 5), marked by downward arrows. These bands are absent in the spectrum recorded immediately after deposition and therefore originate from species formed during the annealing process. Compared to the ν_{Se-H} mode of the isolated PhSeH monomer at 2351 cm⁻¹, the new bands exhibit red shifts of 75 and 61 cm⁻¹, respectively, indicating the formation of hydrogen-bonded complexes. To facilitate assignment of these bands, the stability and simulated ν_{Se-H} mode of five possible (PhSeH)₂ conformers (D1-cis, D2-cis, D1-trans, D2-trans, and D3) were considered in the electronic structure calculations. The calculated stabilisation energies and complexation-induced shifts in ν_{Se-H} are summarised in Table 1. Among the five predicted (PhSeH)₂ structures, D1 and D2 types correspond to Se–H···Se hydrogen-bonded minima on the dimer potential energy surface. D1-cis and D2-cis exhibit shifts of 94 and 90 cm⁻¹, respectively, while D1-trans and D2-trans exhibit shifts of 101 and 100 cm⁻¹, respectively, in agreement with the experimentally observed values. Since the experimental FWHM is approximately 4 cm⁻¹, the individual conformers within each pair are not expected to be resolved separately. Hence, the band at 2276 cm⁻¹ is assigned to the trans pair, D1-trans and D2-trans, while the

band at 2290 cm^{-1} is assigned to the cis pair, D1-cis and D2-cis. The D3 conformer exhibits a considerably smaller calculated red shift of 17 cm^{-1} and is therefore not assigned to the observed bands.

The matrix-isolation IR spectra of PhSeH, $\text{S}(\text{CH}_3)_2$, and $[\text{PhSeH-S}(\text{CH}_3)_2]$ mixture recorded in Ar matrix at 16 K before and after annealing are presented in Figure 5. The spectrum of isolated $\text{S}(\text{CH}_3)_2$ recorded after deposition and annealing show three transitions with very weak intensity associated with $\text{S}(\text{CH}_3)_2$ monomer. The infrared spectrum of the $[\text{PhSeH-S}(\text{CH}_3)_2]$ mixture recorded immediately after deposition is shown in Figure 5. No additional bands are observed in the $\nu_{\text{Se-H}}$ region apart from the monomer of band of PhSeH and weak transitions of $\text{S}(\text{CH}_3)_2$. It indicates that the complex formation is negligible under the initial deposition conditions. Annealing of the matrix results in the appearance of a new band with very weak intensity at 2223 cm^{-1} , marked with an asterisk (*) in Figure 5.

A careful investigation of the IR spectra of PhSeH and $\text{S}(\text{CH}_3)_2$ reveal that the 2223 cm^{-1} band is absent in the monomer spectra of both species and therefore originates from a species formed during the annealing process. The observed band is red-shifted by 128 cm^{-1} compared to the $\nu_{\text{Se-H}}$ mode of the PhSeH monomer, indicating significant weakening of the Se–H bond upon complex formation. Comparison with the simulated vibrational frequencies and the predicted complexation-induced shifts supports assignment of this band to the $\nu_{\text{Se-H}}$ mode of the $\text{PhSeH-S}(\text{CH}_3)_2$ complex.

4. Further Discussion

A comparison of the hydrogen bonding interaction in molecular complexes with other chalcogen atoms, *i.e.* O and S atoms, has been discussed in this section to understand the structure, stability, and hydrogen bond-induced red-shift in the $\nu_{\text{Se-H}}$ mode of $\text{Se-H} \cdots \text{Se}$ and $\text{Se-H} \cdots \text{S}$ hydrogen bond interactions observed in $(\text{PhSeH})_2$ and $[\text{PhSeH-S}(\text{CH}_3)_2]$. $(\text{PhOH})_2$ has been investigated in the gas phase under supersonic jet expansion conditions [51] and in Ar matrix [52]. A red-shift of 146 cm^{-1} was observed in the $\nu_{\text{O-H}}$ mode of $(\text{PhOH})_2$ compared to the PhOH under matrix isolation conditions, whereas a red-shift of 127 cm^{-1} was identified in a supersonic jet experiment. The $(\text{PhOH})_2$ structure is stabilised by the formation of $\text{O-H} \cdots \text{O}$ hydrogen bond interaction without showing any evidence of the $\pi-\pi$ type of interaction, *i.e.*, a structure of open form with respect to the two benzene rings stabilised by hydrogen bond only [51, 52]. The $(\text{PhSH})_2$ was investigated using rotational spectroscopy [53]. Two different types of structures stabilised by $\text{S-H} \cdots \text{S}$ hydrogen bond along with $\pi-\pi$ interactions were identified. The global minimum structure exhibited $\text{S-H} \cdots \text{S}$ hydrogen bond distance of $2.84\text{ \AA}$ and hydrogen bond angle of 134.5° , with a stabilisation energy of 26.9 kJ/mol . The $\text{S-H} \cdots \text{S}$ hydrogen bond interaction was accompanied by $\pi-\pi$ interaction between the aromatic rings, with dispersion interactions making a major contribution to the stabilisation of the dimer. The $\text{S-H} \cdots \text{Se}$ hydrogen-bonded complex was identified in $[\text{H}_2\text{S-Se}(\text{CH}_3)_2]$ complex using FTIR spectroscopy in an N_2 matrix. Formation of the $\text{S-H} \cdots \text{Se}$ hydrogen bond resulted in a large red shift of 146.3 cm^{-1} in the $\nu_{\text{S-H}}$ mode of H_2S , while the calculated binding energy of the binary complex was 15.5 kJ/mol [54].

The $\text{O-H} \cdots \text{S}$ hydrogen-bond interaction has been investigated under supersonic-jet conditions in $[\text{PhOH-S}(\text{CH}_3)_2]$ complex. A red shift of 130 cm^{-1} was observed in the $\nu_{\text{S-H}}$ stretching mode of PhOH [10]. The $\text{O-H} \cdots \text{O}$ and $\text{O-H} \cdots \text{S}$ hydrogen-bonding interactions in p-cresol complexes with diethyl ether (DEE) and diethyl sulfide (DES) have been investigated using IR-UV double-resonance spectroscopy under supersonic-jet conditions. A red shift of 280 cm^{-1} was observed in the $\nu_{\text{O-H}}$ mode of the p-cresol in [p-cresol-DEE] complex, while red shifts ranging from 274 to 292 cm^{-1} were observed for the three conformers of the [p-cresol-DES] complex [11]. A red shift of 127 and 102 cm^{-1} in the $\nu_{\text{O-H}}$ mode of p-cresol was identified in the complexes with H_2O and H_2S , respectively [12]. The $\text{O-H} \cdots \text{Se}$ hydrogen-bonding interaction has been investigated in the $[\text{PhOH-Se}(\text{CH}_3)_2]$ complex using gas-phase IR spectroscopy. A red shift of 240 cm^{-1} was observed in the $\nu_{\text{O-H}}$ mode of PhOH [55].

In the present study, $\text{Se-H} \cdots \text{Se}$ and $\text{Se-H} \cdots \text{S}$ hydrogen bond interactions have been investigated under cryogenic conditions. The two $(\text{PhSeH})_2$ dimers exhibited red shifts of 75 and 61 cm^{-1} in the $\nu_{\text{Se-H}}$, whereas a larger red shift of 128 cm^{-1} was observed in $[\text{PhSeH-Se}(\text{CH}_3)_2]$ complex. The observed red shift of 128 cm^{-1} is comparable to the 130 cm^{-1} red shift in $[\text{PhOH-S}(\text{CH}_3)_2]$ complex.

The $(\text{PhSeH})_2$ exhibits simultaead red shifts of $90\text{-}101\text{ cm}^{-1}$ in the $\nu_{\text{Se-H}}$ stretching mode, whereas $[\text{PhSeH-S}(\text{CH}_3)_2]$ shows a substantially larger red shift of 289 cm^{-1} , despite comparable stabilization energies. The $(\text{PhSeH})_2$ dimers are predominantly stabilised by dispersion, which contributes $61\text{-}63\%$ of the attractive interactions, along with $\text{Se-H} \cdots \text{Se}$ hydrogen bond interaction. In contrast, $[\text{PhSeH-S}(\text{CH}_3)_2]$ has a higher electrostatic contribution ($\sim 41\%$) along with significant charge transfer (84.62 me). This is consistent with the gas-phase IR study of $\text{S-H} \cdots \text{S}$ hydrogen bond between 2-chlorothiophenol and dimethyl sulfide. A large red-shift of $\sim 150\text{ cm}^{-1}$ was

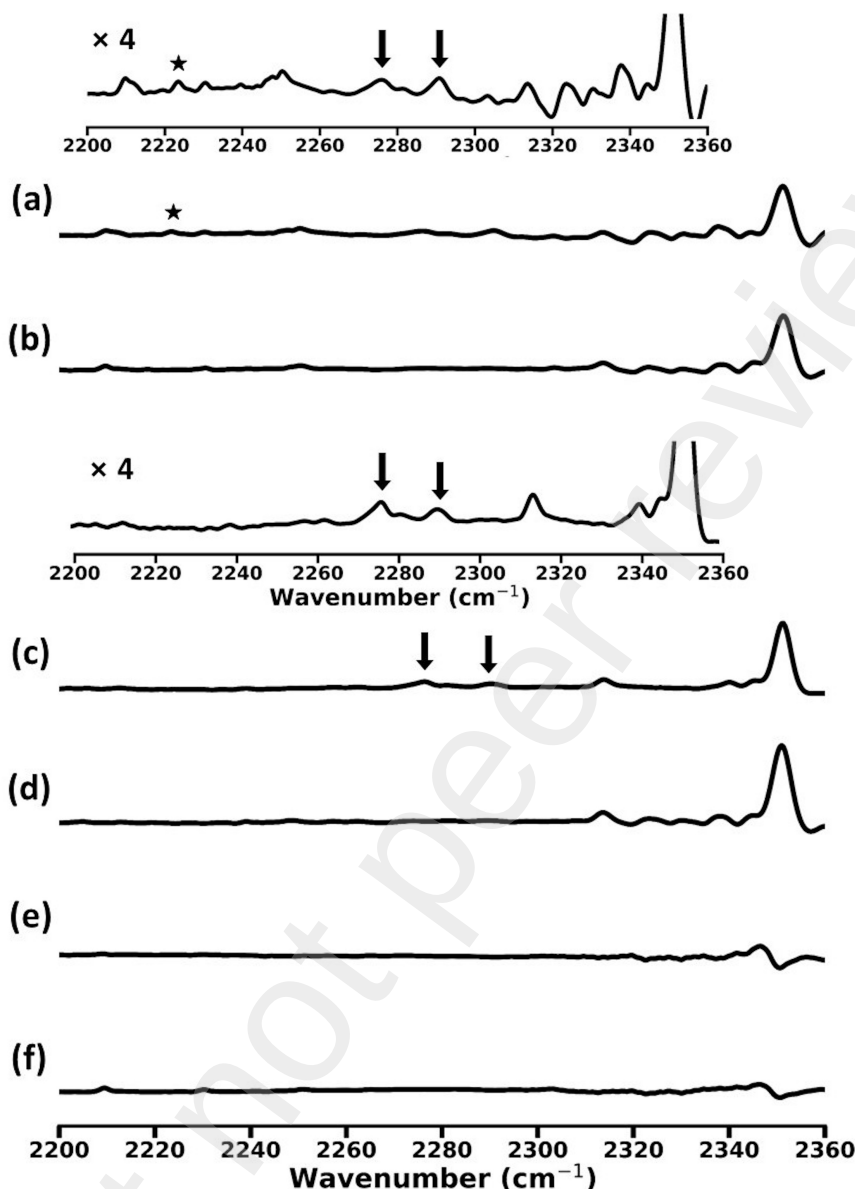


Figure 5: IR spectra recorded in the Ar matrix at 16 K in the $\nu_{\text{Se-H}}$ region. The IR spectra of the $[\text{PhSeH-S(CH}_3)_2]$ mixture after annealing (a), the $[\text{PhSeH-S(CH}_3)_2]$ mixture after deposition (b), PhSeH after annealing (c), PhSeH after deposition (d), $\text{S(CH}_3)_2$ after annealing (e), and $\text{S(CH}_3)_2$ after deposition (f) are shown. The IR transitions assigned to the $(\text{PhSeH})_2$ are marked with a downward arrow ($\downarrow$) and the $[\text{PhSeH-S(CH}_3)_2]$ complex is marked with asterisks ($\star$). The spectra of PhSeH and $[\text{PhSeH-S(CH}_3)_2]$ mixture after annealing are expanded and presented as insets in (a) and (c).

observed in the $\nu_{\text{S-H}}$ mode. The stabilisation of the complex was attributed to significant charge-transfer in addition to electrostatic attraction. The stability of $\text{S-H} \cdots \text{S}$ hydrogen bond in the cis and trans conformers was calculated to be 14.20 and 18.29 kJ/mol, respectively [55].

Overall, the results demonstrate that $\text{Se-H} \cdots \text{Se}$ hydrogen bond in $(\text{PhSeH})_2$ and the $\text{Se-H} \cdots \text{S}$ hydrogen bond in the $[\text{PhSeH-S(CH}_3)_2]$ complex exhibit distinctly different effects on the $\nu_{\text{Se-H}}$ mode. The $\text{Se-H} \cdots \text{Se}$ hydrogen bond in $(\text{PhSeH})_2$ shows smaller red shifts of 61 and 75 cm^{-1} , whereas the $\text{Se-H} \cdots \text{S}$ hydrogen bond in the $[\text{PhSeH-S(CH}_3)_2]$ complex shows a larger red shift of 128 cm^{-1} . The larger red shift in $\text{Se-H} \cdots \text{S}$ hydrogen bond

is consistent with the greater electrostatic and charge-transfer contributions to its stabilisation. The red shift observed for Se–H...S hydrogen bond is comparable to those reported for O–H...S and S–H...S hydrogen bond interaction, whereas the Se–H...Se hydrogen bond interaction shows a smaller red shift.

5. Conclusion

The present study provides the first spectroscopic evidence for Se–H...Se and Se–H...S hydrogen-bond interactions, as revealed by matrix-isolation infrared spectroscopy and supported by electronic-structure calculations. The formation of (PhSeH)₂ and the [PhSeH-S(CH₃)₂] complex was established from characteristic changes in the $\nu_{\text{Se-H}}$ mode. The potential energy surface scan revealed multiple minima corresponding to different conformers of (PhSeH)₂. The lowest-energy structures are stabilised by a combination of Se–H...Se hydrogen bond and $\pi \cdots \pi$ interactions, with four closely lying conformers separated by only 1.33 kJ/mol, whereas other conformers are primarily stabilised by Se–H... π interactions. The experimentally observed red shifts of 75 and 61 cm^{−1} for the $\nu_{\text{Se-H}}$ mode provide spectroscopic signatures of the trans and cis dimer pairs, respectively, which are stabilised by Se–H...Se hydrogen bond interactions. The [PhSeH-S(CH₃)₂] complex is stabilised by a Se–H...S hydrogen bond, as indicated by a stabilisation energy of 24.19 kJ/mol and a substantial red shift of 128 cm^{−1} in the Ar matrix. The nature of these interactions is further supported by quantum-chemical analyses. The presence of bond critical points along the Se–H...Se and Se–H...S bond paths confirms the existence of hydrogen bond interactions, while the corresponding electron-density characteristics indicate that the interactions are predominantly non-covalent in nature. NBO analysis shows greater charge transfer for the Se–H...S interaction than for the Se–H...Se interaction. The EDA analysis reveals that (PhSeH)₂ is predominantly dispersion-driven, whereas the [PhSeH-S(CH₃)₂] complex possesses relatively larger electrostatic and induction contributions. Furthermore, the spectroscopic and theoretical results demonstrate that selenium can act as a hydrogen-bond donor through the Se–H group and that Se–H...Se and Se–H...S interactions can contribute significantly to the structure and stabilisation of molecular complexes.

Acknowledgement

The authors thank the computer centre at BITS-Pilani, Pilani Campus, where all calculations were performed, for providing time in the High-Performance Computational Facility.

CRedit authorship contribution statement

Sonika Sangwan: Methodology, Investigation, Formal analysis, Writing - Original Draft. **Heena Charaya:** Methodology, Investigation, Formal analysis, Writing - Review & Editing. **Shamik Chakraborty:** Conceptualisation, Methodology, Formal analysis, Supervision, Writing - Review & Editing.

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